

Prototypical Representation Learning for Multi-Site Domain Generalization in Schizophrenia Diagnosis

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Abstract— Brain functional networks (BFNs) derived from multi-site fMRI data have been widely used in the diagnosis of psychiatric disorders such as schizophrenia (SZ). However, site-induced distribution shifts pose a major challenge to the generalization of classification models based on BFNs. While domain generalization (DG) seeks to address this challenge without access to target domain data, most methods make the idealized assumption that each class exhibits consistent structures across sites. The inherent diversity within each class often violates this assumption, resulting in poor generalization to unseen domains. To address this issue, we proposed a DG framework based on a transformer encoder combined with prototype learning. We first employed a transformer encoder to capture global dependencies among brain regions and extract discriminative subject-level representations from BFNs. A site-independence module with Hilbert-Schmidt Independence Criterion (HSIC) regularization was then applied to enforce site-invariant feature learning. Projected features were softly assigned to multiple prototypes via Sinkhorn matching, with prototypes updated via exponential moving average (EMA), and the maximum likelihood estimation (MLE) loss further refining feature-to-prototype matching probabilities. Contrastive and alignment losses were applied to promote inter-class separability and intra-class consistency, respectively. Experimental results showed that our method outperformed the other 7 DG, 6 domain adaptation (DA), 6 multi-site and 6 state-of-the-art methods across two independent SZ datasets ($88.89\% \pm 2.22\%$ and $86.05\% \pm 1.64\%$). Ablation results highlighted the contributions of the MLE, contrastive, and alignment constraints to the performance improvement. Furthermore, the identified discriminative temporal regions provided insights into the dysfunctional neural-mechanisms in SZ.

Index Terms— Prototypical representation learning, multi-site brain functional networks, domain generalization, diagnostic classification.

I. INTRODUCTION

SCHIZOPHRENIA (SZ) is a chronic mental disorder that affects a large population worldwide. Accurate and early diagnosis is essential for timely intervention and effective treatment. Resting-state functional magnetic resonance imaging (fMRI) has become a promising tool to aid in the diagnosis of SZ, as it allows the construction of brain functional networks (BFNs) revealing abnormal connectivity patterns reflecting the neuropathology of SZ [1]. In recent years, numerous studies have increasingly focused on applying machine learning methods to BFNs for automated diagnostic classification.

The application of BFN-based diagnostic models is hindered by distributional discrepancies across sites. Although models generally perform well when trained and tested on data from the same site, variations in scanners, imaging protocols, and participant demographics can lead to substantial distributional shifts [2]. As a result, a model trained on BFNs from one site may struggle to generalize to BFNs from other sites, limiting its reliability on unseen data [3]. To address this challenge, domain adaptation (DA) and domain generalization (DG) have been explored. DA methods are the most widely used in BFN analysis, aiming to bridge cross-site distribution discrepancies by aligning data from the source domains and the target domains to approximate a shared distribution. They leverage labeled data from the source domains to train a model and directly utilize the learned model to predict the labels of the unlabeled data from the target domains [4]. Kunda et al. [5] proposed a DA method that utilized tangent Pearson embedding for second-order feature extraction, and applied Hilbert-Schmidt Independence Criterion (HSIC)-based regularization to minimize site dependence in multi-site BFNs. Fang et al. [6] proposed a discrepancy-based unsupervised DA framework that integrated a maximum mean discrepancy (MMD) module for cross-domain feature alignment, along with an attention-guided graph convolutional network (GCN) for learning domain-invariant representations. Li et al. [7] developed a privacy-preserving federated learning framework for multi-site BFN analysis, using adversarial domain alignment to learn domain-invariant features and enhance

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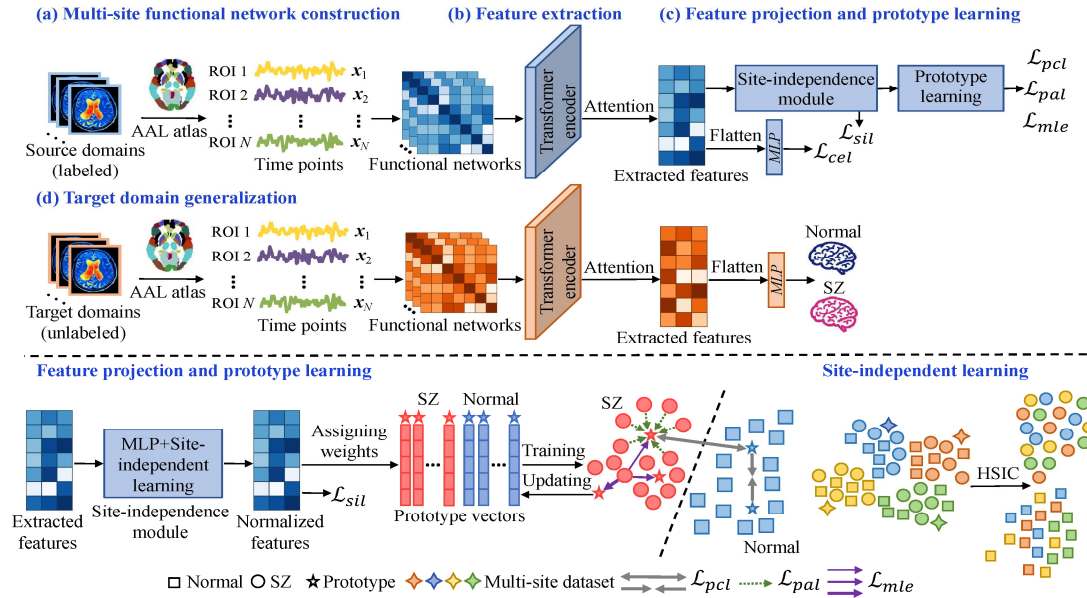


Fig. 1. Flowchart of the proposed DG framework. (a) BFNs were constructed from resting-state fMRI across multiple sites and were used as input to the model. (b) A transformer encoder extracted subject-level embeddings by modeling global inter-regional dependencies. (c) The extracted features were projected with HSIC regularization to encourage site-invariant representations. A prototype learning module assigned projected features to class-specific prototypes via the Sinkhorn-Knopp algorithm. The resulting soft assignments guided prototype updates through EMA, while the MLE loss subsequently optimized the feature-to-prototype matching probabilities in the probabilistic space. To further enhance generalizability, contrastive and alignment losses were introduced to promote inter-class separation and intra-class compactness, respectively. (d) The model was directly evaluated on unseen target domains for classification.

cross-site classification. While DA methods are widely employed for multi-site BFN analysis, their reliance on target domain data is impractical because testing data is inherently unavailable in real-world applications [8]. In contrast, DG methods are more suitable as they learn invariant representations from multiple source domains, making them generalizable to an unseen target domain [9].

Several studies have leveraged DG methods to tackle the challenges of multi-site BFN heterogeneity. Lee et al. [10] proposed a meta-learning framework that adaptively calibrated site-specific features into site-invariant representations via a novel modulation mechanism, improving the generalizability of diagnostic models to unseen sites in BFN analysis. Qiu et al. [11] introduced a meta-learning-based DG framework that simultaneously enhanced the interpretability and generalizability of GCNs by incorporating explanatory regularization, enabling the learning of cross-site consistent representations. Su et al. [12] proposed a meta-learning framework that integrated meta batch normalization and distance-based constraints to capture spatial-temporal patterns from BFNs and enforce feature consistency across domains, thereby improving the performance on unseen sites. However, DG methods must be carefully tailored to diagnostic classification tasks to ensure that reliable and robust decision boundaries can be learned even in the absence of target domain data [13]. In addition, most current DG methods assume that each class exhibits consistent structural patterns across sites, an overly idealized assumption that overlooks the inherent diversity within classes and subtle site-induced variations [14]. Modeling intra-class diversity enables the model to capture a broader range of valid class-specific variations, which is critical for improving generalization to unseen domains with structural deviations.

To address these challenges, we proposed a DG framework that integrated prototype learning with site-invariant representation to jointly enhance multi-site BFN classification. Specifically, a transformer encoder was employed to capture global inter-regional dependencies from BFNs, generating discriminative subject-level embeddings. To mitigate site-specific biases, a site-independence module with an HSIC-regularized projection was introduced to encourage site-invariant representations. A prototype learning module was designed to assign projected features to class-specific prototypes via the Sinkhorn-Knopp algorithm. Prototypes were updated using the exponential moving average (EMA) method, while the maximum likelihood estimation (MLE) loss optimized the matching probabilities between projected features and prototypes in the probabilistic space. In addition, contrastive and alignment losses were employed to enhance inter-class separability and intra-class compactness. The proposed method was validated in the diagnosis of SZ, outperforming 25 comparative methods on two datasets.

II. METHODS AND MATERIALS

A. Overview

Fig. 1 gives the overview of our method, includes three main components: (a) BFNs were constructed from resting-state fMRI data across multiple sites. (b) A transformer encoder was employed to capture global inter-regional dependencies within each BFN, generating subject-level embeddings that reflect brain-wide connectivity patterns. (c) The extracted features were passed through a site-independence module, in which an MLP-based projection is regularized by HSIC to enhance site invariance. A prototype learning module assigned projected features to class-specific prototypes via soft assignments, with

prototype updates performed using EMA, guided by the MLE loss to reflect probabilistic associations. Contrastive and alignment losses were further employed to promote inter-class separability and intra-class consistency, respectively. (d) The trained model was evaluated directly on the unlabeled target domain, to assess its generalization to unseen sites.

B. Datasets

The fMRI data used in this study was acquired from Function Biomedical Informatics Research Network (FBIRN) phase datasets [15] and Bipolar-Schizophrenia Network for Intermediate Phenotypes (BSNIP) datasets [16]. FBIRN, including 161 SZs and 157 normal controls (NCs) from seven sites, matched for age, gender, handedness and race distributions, was used as validation dataset 1. BSNIP, comprising 133 SZs and 160 NCs from four sites, was used as validation dataset 2. All participants provided written informed consent at their respective sites, and the original data collection procedures were approved by the local institutional review boards of all participating institutions in accordance with the Declaration of Helsinki. Detailed site and fMRI preprocessing information can be found in **Supplementary Datasets**.

C. Construction of BFNs and Feature Extraction

The anatomical automatic labeling (AAL) atlas (90 brain regions, without cerebellum) [17] was used as the template to construct BFNs. To better characterize inter-regional dependencies and learn more discriminative representations, we employed a transformer encoder to process the BFNs of each subject [18]. By aggregating and updating the global region-wise representations based on attention weights, the transformer enables a deeper understanding of the BFN structure. Specifically, each row of the BFN $\mathbf{X} \in \mathbb{R}^{N \times N}$ serves as a feature vector for a brain region's connectivity pattern. The self-attention mechanism calculates attention scores between regions and aggregates their representations accordingly. Define $\mathbf{z}^l \in \mathbb{R}^{N \times d}$ as the matrix of region-wise representations at the l -th layer for a BFN, d denotes the feature dimension of each region, which corresponds to the connectivity profile dimension in the input layer and becomes a latent embedding dimension in higher layers. The self-attention mechanism in the h -th head can be expressed as:

$$\mathbf{z}^{h,l} = \mathbf{A}^{h,l} \mathbf{z}^l \mathbf{W}_V^{h,l} \quad (1)$$

where $\mathbf{A}^{h,l} = \text{softmax}\left(\frac{\mathbf{z}^l \mathbf{W}_Q^{h,l} (\mathbf{z}^l \mathbf{W}_K^{h,l})^T}{\sqrt{d_k}}\right) \in \mathbb{R}^{N \times N}$ is the matrix

of attention scores among all regions in the l -th layer of the h -th head, $\mathbf{W}_Q^{h,l}$, $\mathbf{W}_K^{h,l}$, $\mathbf{W}_V^{h,l} \in \mathbb{R}^{d \times d_h}$ is the matrix of learnable linear transformations applied to the region-wise features, and d_h denotes the hidden dimension of each attention head, i.e., the column dimension of $\mathbf{z}^l \mathbf{W}_K^{h,l}$.

To better capture comprehensive information about the BFN, a multi-head self-attention mechanism was employed for parallel computation. The input is processed in parallel by H distinct self-attention modules, producing H output matrices $\mathbf{Z} \in \mathbb{R}^{N \times d}$, which are then aggregated as follows:

$$\mathbf{Z}^l = \text{LN}\left(\mathcal{W}(\mathcal{C}(\mathbf{Z}^{1,l}, \mathbf{Z}^{2,l}, \dots, \mathbf{Z}^{H,l}))\right) \quad (2)$$

where $\mathcal{C}$ is the concatenation operation, $\mathcal{W}$ denotes linear transform layer, and LN denotes layer normalization. After feature learning via self-attention, a new feature representation $\mathbf{Z}$ was obtained by stacking multiple transformer layers. Then, an attention-based aggregation mechanism was employed to obtain a compact subject-level representation from the transformer-encoded brain region features. Specifically, attention scores were computed via a linear layer and normalized using a *softmax* function to produce region-wise weights, which were then used to perform a weighted sum over all brain regions. The obtained feature vectors for each subject were subsequently used for prototype learning and classification.

D. Feature Projection and Prototype Learning

The feature representations were learned through joint optimization of HSIC regularization and prototype learning, promoting both site invariance and class discriminability. Specifically, a multi-layer perceptron (MLP) was applied to introduce non-linear transformations and capture more compact and discriminative representations, making class-related information more prominent while reducing the influence of site-specific variations. Then, a HSIC regularization term was introduced to reduce site-specific biases and promote site-invariant feature representations. We assume that the Q subjects are collected from S source domains, with site information represented by one-hot site vectors $\{\mathbf{s}_i\}_{i=1}^Q \in \mathbb{R}^{Q \times S}$, where $\mathbf{s}_i = [s_{i,1}, s_{i,2}, \dots, s_{i,S}]^T \in \mathbb{R}^S$ and $s_{i,j}=1$ if subject i from site j . HSIC is a non-parametric statistics to measure the dependence between two multi-variate random variables, and a larger HSIC value means stronger dependence [19]. Hence, the site-independence module aims to minimize the empirical HSIC value between the MLP-projected feature representations $\mathbf{z}_i$ and the one-hot site vectors $\mathbf{s}_i$:

$$\mathcal{L}_{sil} = \frac{1}{(Q-1)^2} \text{tr}(\mathbf{Z} \mathbf{U} \mathbf{S} \mathbf{U}) \quad (3)$$

where $\mathbf{U} \in \mathbb{R}^{Q \times Q} = \mathbf{I} - \frac{1}{Q} \mathbf{1} \mathbf{1}^T$, $\mathbf{1} \in \mathbb{R}^Q$ denotes a column vector of ones and $\mathbf{1} \mathbf{1}^T$ forms the all-ones matrix used for mean-centering the kernel features. $\mathbf{Z}, \mathbf{S} \in \mathbb{R}^{Q \times Q}$ with $Z_{i,j} = \varphi_z(\mathbf{z}_i, \mathbf{z}_j)$, $S_{i,j} = \varphi_s(\mathbf{s}_i, \mathbf{s}_j)$, $\varphi_z(\cdot, \cdot)$ and $\varphi_s(\cdot, \cdot)$ are two linear kernel functions. Minimizing $\mathcal{L}_{sil}$ encourages the learned features to reduce dependence on site-specific variations, thereby promoting the preservation of class-discriminative information.

Prototype learning aims to learn a representative vector (prototype) for each class, thereby enabling more discriminative and robust classification [20]. In this study, we applied prototype learning to the projected features to reflect intra-class diversity by maintaining multiple prototypes per category. Each prototype can be interpreted as a latent anchor in the transformer-encoded feature space that summarizes the characteristic connectivity representations shared by subjects within the same clinical group, capturing the central topological and interactional traits of their BFNs. In contrast to prior studies [21] that adopted a single class-conditional prototype to represent all samples of a class, this design often fails to capture

intra-class diversity, leading to limited representational capacity, non-compact embeddings, and confusion between in-distribution and out-of-distribution samples. Inspired by Lu et al. [22], we therefore define multiple prototypes per class to better model intra-class diversity.

First, K prototypes were defined for each class c . To initialize these prototypes, we sampled from a standard Gaussian distribution and applied L_2 normalization to ensure they lie on the unit hypersphere. Then, the soft assignment weights of each sample to the prototypes were computed using the Sinkhorn-Knopp algorithm [23]. Specifically, B denotes the number of samples in the batch. The soft assignment weights $\mathbf{W}_c \in \mathbb{R}^{K \times B}$ were obtained by optimizing the following objective that aligns each prototype with its most similar samples:

$$\max_{\mathbf{W}_c \in \mathcal{V}} \text{tr}(\mathbf{W}_c^T \mathbf{P}_c^T \mathbf{Z}_c) + \epsilon R(\mathbf{W}_c) \quad (4)$$

where $\mathbf{P}_c \in \mathbb{R}^{K \times d}$ represents the prototypes corresponding to class c , $\mathbf{Z}_c \in \mathbb{R}^{B \times d}$ denotes the feature representations belonging to class c . $R(\cdot)$ is an entropy-based regularization function and $R(\mathbf{W}_c) = -\sum_{i,j} W_{ij} \log W_{ij}$, ϵ is the regularization coefficient. The feasible set $\mathcal{V}$ is defined as the transportation polytope, i.e., the set of doubly stochastic matrices representing valid transport plans between the prototype and sample distributions: $\mathcal{V} = \{\mathbf{W} \in \mathbb{R}_+^{K \times B} | \mathbf{W} \mathbf{1}_B = \frac{1}{K} \mathbf{1}_K, \mathbf{W}^T \mathbf{1}_K = \frac{1}{B} \mathbf{1}_B\}$, where $\mathbf{1}_K$ and $\mathbf{1}_B$ denote the vector of ones in dimension of K and B , respectively. The above optimization problem is efficiently solved using the Sinkhorn-Knopp algorithm, which provides a numerically stable and differentiable approximation to the optimal transport plan.

Intuitively, this procedure balances assignments between prototypes and samples, producing a smooth transport plan that softly aligns the prototype and feature distributions. After convergence, the resulting transport plan of Eq. (4) is given by:

$$\mathbf{W}_c = \text{diag}(\mathbf{u}) \exp\left(\frac{\mathbf{P}_c^T \mathbf{Z}_c}{\epsilon}\right) \text{diag}(\mathbf{v}) \quad (5)$$

which corresponds to a normalized exponential matrix representing the final soft assignment weights.

After obtaining the soft assignment weights $\mathbf{W}_c$, the prototypes were updated using the EMA method based on the weighted sample features assigned to each prototype. This method updates prototypes iteratively, maintaining stability in the assignment process. The update rule is:

$$\mathbf{P}_c \leftarrow \text{Normalize}(\alpha \mathbf{P}_c + (1 - \alpha) \mathbf{W}_c^T \mathbf{Z}_c) \quad (6)$$

where α is a momentum coefficient that balances historical information and adaptability. A larger α makes the update smoother and more stable, while a smaller α allows quicker adaptation to the data. Following the prototype update, prototypes are renormalized onto the unit hypersphere to ensure consistent and interpretable distance metrics.

E. Optimization Strategy

With the updated prototypes, we proceed to calculate four downstream losses based on their interactions with sample representations. These losses were then combined, and

backpropagation was used to update the model parameters, while the prototypes continued to be updated via EMA for stability. First, the classification loss is the standard cross-entropy loss, which enforces that the classifier weights for each class are used to assess whether the updated sample representation $\mathbf{z}_i$ belongs to the corresponding class. This term ensures that the classifier output for each sample closely matches the ground-truth labels:

$$\mathcal{L}_{cel} = -\frac{1}{Q} \sum_{i=1}^Q \log \frac{\exp(\mathbf{w}_{y_i}^T \mathbf{z}_i)}{\sum_{c=1}^C \exp(\mathbf{w}_c^T \mathbf{z}_i)} \quad (7)$$

where $\mathbf{w}_{y_i} \in \mathbb{R}^d$ denotes the weight vector of the classifier corresponding to the ground-truth class label y_i .

Beyond classification, we introduce an MLE objective to learn probabilistic feature-prototype matching by maximizing likelihood with respect to same-class prototypes while suppressing others. The resulting Sinkhorn-based soft assignments guide EMA prototype updates. The optimization is formulated as the negative log-likelihood:

$$\mathcal{L}_{mle} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\sum_{k=1}^K \mathbf{w}_{i,k}^{y_i} \exp(\mathbf{P}_k^{y_i T} \mathbf{z}_i / \tau)}{\sum_{c=1}^C \sum_{k'=1}^{K'} \mathbf{w}_{i,k'}^c \exp(\mathbf{P}_{k'}^c T \mathbf{z}_i / \tau)} \quad (8)$$

where τ is a temperature parameter controlling the concentration of similarities. This loss enables flexible and expressive prototype learning by optimizing soft assignments to probabilistically match features with class-level prototypes.

To further enhance the discriminability and separability of the prototypes themselves, we introduced a prototype contrastive loss [24] that directly operated in the prototype space. This term encourages same-class prototypes to group closely together and pushes apart prototypes from different classes, as defined below:

$$\mathcal{L}_{pcl} = -\frac{1}{K} \sum_{i=1}^K (\sum_{j=1}^K M_{ij} S_{ij} - \log \sum_{j=1}^K \exp(S_{ij})(1 - \delta_{ij})) \quad (9)$$

where $S_{ij} = \frac{\mathbf{P}_i^T \mathbf{P}_j}{\tau}$ is the scaled cosine similarity between prototypes i and j , scaled by the temperature parameter τ . δ_{ij} represents the Kronecker delta, acting as an indicator that equals 1 when $i = j$ and 0 otherwise. $\mathbf{M} \in \mathbb{R}^{K \times K}$ is a binary mask matrix to identify pairs of prototypes that belong to the same class, and remove self-similarities using the identity matrix $\mathbf{I} \in \mathbb{R}^{K \times K}$. The positive term $\sum_{j=1}^K M_{ij} S_{ij}$ is computed as the row-normalized average similarity to all other same-class prototypes (excluding itself). In contrast, the negative term $\log \sum_{j=1}^K \exp(S_{ij})(1 - \delta_{ij})$ aggregates the log-sum-exp of similarities to all other prototypes in the set, regardless of class, effectively treating all non-self prototypes as negatives.

While $\mathcal{L}_{mle}$ and $\mathcal{L}_{pcl}$ enhance global structure and inter-class separability through probabilistic alignment and prototype dispersion, they overlook instance-level guidance, which is essential for refining feature-prototype matching and ensuring consistent prototype association at the individual sample level. To address this, we introduced a prototype alignment loss that explicitly supervised the sample-prototype correspondence within each class. This promotes intra-class consistency while enhancing discriminability and stabilizing prototype learning. It can be expressed as:

$$\mathcal{L}_{pat} = \frac{1}{C} \sum_{c=1}^C \frac{1}{|B_c|} \sum_{i \in B_c} \|\mathbf{z}_i - \hat{\mathbf{p}}_i\|_2^2 \quad (10)$$

TABLE I

CLASSIFICATION PERFORMANCE IN COMPARISON WITH DIFFERENT DG METHODS. * INDICATES THAT THE IMPROVEMENT OF OUR METHOD OVER THE SECOND-BEST METHOD IS STATISTICALLY SIGNIFICANT (PAIRED T-TEST, $*p < 0.05$)

Cohort	FBIRN				BSNIP			
Method	ACC	SEN	SPE	AUC	ACC	SEN	SPE	AUC
MMDG[10]	0.8372±0.0407	0.8174±0.0787	0.8631±0.1099	0.8762±0.0657	0.7571±0.0367	0.7813±0.1593	0.7174±0.1237	0.8319±0.0161
MDGEGR[25]	0.7903±0.0186	0.8187±0.0701	0.7638±0.0964	0.8534±0.0237	0.7878±0.0259	0.8112±0.0922	0.7622±0.0589	0.8306±0.0300
VMLDG[26]	0.8192±0.0161	0.8117±0.0876	0.8259±0.0898	0.8188±0.0140	0.8279±0.0127	0.8375±0.0140	0.8000±0.0407	0.8188±0.0198
TDG[27]	0.8266±0.0186	0.8857±0.0261	0.7750±0.0228	0.8865±0.0121	0.7393±0.0059	0.5987±0.1028	0.8115±0.0539	0.7864±0.0180
IDGM[28]	0.8400±0.0099	0.7714±0.0398	0.9000±0.0228	0.8357±0.0114	0.8148±0.0150	0.7758±0.0312	0.8322±0.0178	0.8040±0.0174
IIB[29]	0.8311±0.0122	0.8190±0.0621	0.8417±0.0543	0.8304±0.0128	0.7664±0.0166	0.7163±0.1323	0.8182±0.0963	0.7672±0.0385
SGM[30]	0.8334±0.0119	0.8371±0.1037	0.8256±0.1071	0.8314±0.0104	0.8276±0.0165	0.7583±0.0513	0.8587±0.0409	0.8085±0.0101
Ours	0.8889±0.0222*	0.9143±0.0976	0.8667±0.0457	0.9433±0.0295*	0.8605±0.0164*	0.8750±0.0221*	0.8182±0.0643	0.8392±0.0192

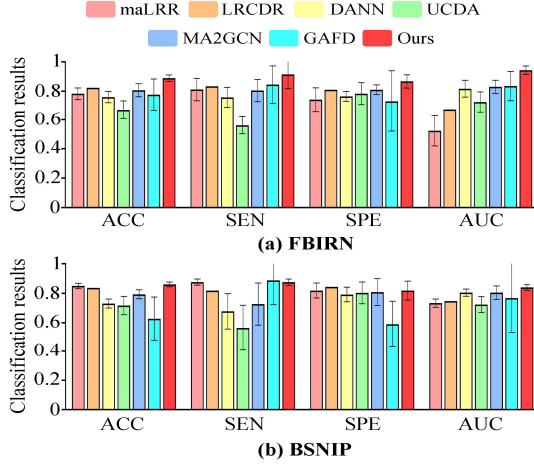


Fig. 2. Classification performance in comparison with different DA methods on two datasets. (a) FBIRN, (b) BSNIP.

where $\mathcal{B}_c$ represents the set of sample indices in the batch that belong to class c . $\hat{\mathbf{p}}_i \in \mathbb{R}^d$ represents the prototype in $\mathcal{P}_c$ that has the highest inner product with the sample embedding. The squared Euclidean distance directly enforces intra-class compactness, reducing within-class variance. It is worth noting that both $\mathcal{L}_{mle}$ and $\mathcal{L}_{pal}$ aim to associate features with prototypes. However, $\mathcal{L}_{mle}$ operates in the probabilistic space by optimizing soft assignments, encouraging features to move toward class-consistent prototypes in a distribution-aware manner, while $\mathcal{L}_{pal}$ enforces direct deterministic alignment between features and prototypes. Their complementary roles lie in $\mathcal{L}_{mle}$ offering global probabilistic guidance, while $\mathcal{L}_{pal}$ enhances local feature-level compactness.

Thus, the overall training objective can be formally defined as:

$$\mathcal{L}_{overall} = \mathcal{L}_{cel} + \mathcal{L}_{mle} + \lambda_1 \mathcal{L}_{pcl} + \lambda_2 \mathcal{L}_{pal} + \lambda_3 \mathcal{L}_{sil} \quad (11)$$

We validated the effectiveness of our method on DG tasks, where the model was trained using labeled data from multiple source domains. Without any target-specific fine-tuning, the model was directly evaluated on an unseen target domain, thereby assessing its ability to generalize to new, unseen data distributions.

III. RESULTS

A. Experimental Settings

We followed the strategy of Qiu et al. [25] by randomly selecting one site as the target domain and treating the remaining sites as source domains. This procedure was repeated 5 times to compute an average performance that robustly evaluates the DG capability. We adopted a 2-layer,

4-head transformer encoder to map BFNs into 256-dimensional embeddings, followed by attention pooling and a projector that produced 128-dimensional normalized representations. The projector is implemented as a two-layer MLP that maps the 256-dimensional encoder features to a 128-dimensional space, with a ReLU activation applied after the first layer. A total of 50 class-aware prototypes were initialized in this space and updated via momentum ($\alpha = 0.99$), with normalization onto the unit hypersphere after each step. In $\mathcal{L}_{mle}$, the similarity scores were scaled by a temperature $\tau = 0.1$. The model was trained using the Adam optimizer with a learning rate of $1e-3$, and all experiments were conducted for 50 epochs. To evaluate the performance of different methods, four classification metrics were utilized, including accuracy (ACC), sensitivity (SEN), specificity (SPE), and area-under-curve (AUC). Higher values for these metrics indicate better classification performance.

B. Comparisons With Different DG Methods

To verify the effectiveness of our method, we compared it with different DG methods. We compared our method with two representative BFN-specific DG methods: (1) Meta-modulation network for DG (MMDG) [10], (2) meta-learning DG with explainability-generalizable regularizations (MDGEGR) [25], and five mainstream DG methods: (3) virtual meta-learning for DG (VMLDG) [26], (4) theoretical DG (TDG) [27], (5) inter-domain gradient matching (IDGM) [28], (6) invariant information bottleneck (IIB) [29], and (7) smoothed gradient masking (SGM) [30]. For fair comparison, all hyper-parameters of all comparison methods were selected according to their original settings. For each run, one site was randomly selected as the target domain while the remaining sites served as source domains. This process was repeated five times to report the average performance.

The classification performance in distinguishing SZ and NC was displayed in TABLE I. Our method consistently achieved the best performance in terms of ACC, SEN, and AUC on both datasets, exceeding the ACC of all comparison methods by at least 4.89% on the FBIRN and 3.26% on the BSNIP. Furthermore, our method surpassed meta-learning-based methods such as MMDG, MDGEGR, VMLDG, and IDGM. This is mainly because meta-learning methods involve complex nested optimization that may hinder stability and generalization, whereas our method improves cross-site robustness by modeling class-wise diversity through prototype-based representation learning.

C. Comparisons With Different DA Methods

To comprehensively evaluate our method, we also compared

TABLE II

CLASSIFICATION PERFORMANCE IN COMPARISON WITH DIFFERENT MULTI-SITE RELATED METHODS. * INDICATES THAT THE IMPROVEMENT OF OUR METHOD OVER THE SECOND-BEST METHOD IS STATISTICALLY SIGNIFICANT (WELCH'S TWO-SAMPLE T-TEST, ** $p < 0.01$, * $p < 0.05$)

Cohort	FBIRN				BSNIP			
Method	ACC	SEN	SPE	AUC	ACC	SEN	SPE	AUC
ComBat[36]	0.8045±0.0471	0.7898±0.0786	0.8195±0.0626	0.8683±0.0357	0.7818±0.0625	0.7428±0.1313	0.8150±0.0631	0.8478±0.0517
MHAHGEL[37]	0.5560±0.0309	0.5719±0.0411	0.5468±0.0580	0.5710±0.0301	0.5529±0.0100	0.5321±0.0419	0.5712±0.0485	0.5699±0.0135
RMDRL[38]	0.7451±0.0276	0.6728±0.1062	0.8108±0.0734	0.8123±0.0375	0.7645±0.0137	0.6986±0.0767	0.8303±0.0765	0.8763±0.0293
SAENet[39]	0.7337±0.0525	0.6413±0.1946	0.8314±0.1109	0.8291±0.0451	0.7716±0.0114	0.6392±0.1323	0.8262±0.0596	0.8052±0.0128
CCLM[40]	0.7806±0.1412	0.8051±0.2337	0.7397±0.1900	0.8033±0.1195	0.8525±0.0561	0.8372±0.0832	0.8820±0.2173	0.8905±0.0803
DMSGCN[41]	0.6477±0.1157	0.6252±0.2864	0.7020±0.2199	0.7357±0.1125	0.6473±0.1030	0.4516±0.2843	0.8133±0.1207	0.6615±0.1430
Ours	0.8889±0.0222**	0.9143±0.0976**	0.8667±0.0457	0.9433±0.0295**	0.8605±0.0164*	0.8750±0.0221	0.8182±0.0643	0.8392±0.0192

TABLE III

CLASSIFICATION PERFORMANCE IN COMPARISON WITH STATE-OF-THE-ART METHODS

Cohort	FBIRN				BSNIP			
Method	ACC	SEN	SPE	AUC	ACC	SEN	SPE	AUC
SSGSR[42]	0.6930±0.0548	0.6873±0.0901	0.7168±0.0830	0.5168±0.0768	0.6709±0.0534	0.7286±0.0848	0.6045±0.0859	0.6558±0.0679
SRLTD[43]	0.6987±0.0595	0.6954±0.0972	0.7055±0.0930	0.4892±0.0868	0.5631±0.0626	0.6249±0.0939	0.4981±0.1071	0.5499±0.0695
MDGL[44]	0.8204±0.0282	0.7531±0.0256	0.6961±0.0369	0.9254±0.0149	0.6950±0.0371	0.5874±0.1756	0.7699±0.0559	0.7599±0.0742
BCNN[45]	0.8304±0.0112	0.7949±0.0316	0.8626±0.0207	0.8643±0.0141	0.8361±0.0312	0.7942±0.1093	0.8681±0.0651	0.8863±0.0326
VATCNN[46]	0.7948±0.0855	0.8544±0.0712	0.7251±0.0939	0.8767±0.0963	0.6161±0.0605	0.8036±0.0803	0.5368±0.1835	0.6463±0.1282
MGCN[47]	0.7352±0.0099	0.7368±0.0180	0.7348±0.0217	0.8223±0.0075	0.7454±0.0209	0.6853±0.0271	0.7932±0.0270	0.8172±0.0204
Ours	0.8889±0.0222	0.9143±0.0976	0.8667±0.0457	0.9433±0.0295	0.8605±0.0164	0.8750±0.0221	0.8182±0.0643	0.8392±0.0192

it with several DA methods, including: (1) Multi-site adaption framework via low-rank representation decomposition (maLRR) [31], (2) low-rank and class discriminative representation (LRCDDR) [32], (3) domain adversarial neural network (DANN) [33], (4) unsupervised cross-domain adaptation (UCDA) [6], (5) multi-site adaptation with attention GCN (MA2GCN) [34], (6) graph augmentation guided federated distillation (GAFD) [35].

The classification performance of different methods was displayed in **Fig. 2** and was reported as the average of five repeated experiments, each with a randomly selected target domain, which is further split into training and testing subsets using 5-fold cross-validation and the remaining domains used as source domains. Our method achieved the best performance on all four metrics on the FBIRN and outperformed all comparison methods in terms of ACC and AUC on the BSNIP. Unlike DA, which relies on target domain data, DG enhances robustness and generalizability, making it better suited for real-world applications where target data is limited.

D. Comparisons With Different Multi-Site Related Methods

Beyond DG and DA methods, various other methods have also been proposed to tackle the challenges posed by inter-site heterogeneity, including: (1) Combat harmonization method (ComBat) [36], (2) multi-view hyperedge-aware hypergraph embedding learning (MHAHGEL) [37], (3) Riemannian manifold-based disentangled representation learning (RMDRL) [38], (4) sparse autoencoder network (SAENet) [39], (5) consistent connectome landscape mining (CCLM) [40], (6) dynamic multi-site GCN (DMSGCN) [41].

As shown in **TABLE II**, compared with these methods, our method achieved the best performance on all four metrics in FBIRN and achieved the best ACC and SEN in BSNIP. Unlike methods relying on explicit harmonization (ComBat), complex graph construction (MHAHGEL, DMSGCN), geometry-aware disentanglement (RMDRL), or task-specific representation learning (SAENet, CCLM), our method focuses on learning a unified feature space that is inherently robust to site shifts and class-discriminative. Instead of manually correcting for site effects or constructing complex multi-site modeling

architectures, our method implicitly promotes distributional invariance and class-level alignment, resulting in more stable and generalizable performance across unseen domains.

E. Comparisons With State-of-The-Art Methods

We also compared our method with several state-of-the-art methods, including: (1) Strength and similarity guided group sparse representation (SSGSR) [42], (2) sparse representation with latent temporal dependency (SRLTD) [43], (3) multi-scale dynamic graph learning (MDGL) [44], (4) brain network convolutional neural network (BCNN) [45], (5) virtual adversarial training CNN (VATCNN) [46], and (6) modularity-constrained GCN (MGCN) [47].

As shown in **Table III**, our method achieved superior performance over 6 state-of-the-art methods. Compared to methods based on mainstream deep learning architectures such as CNN-based methods like BCNN and VATCNN, GCN-based methods like MGCN, and transformer-based methods like MDGL, our method achieves the highest accuracy on both datasets, surpassing the second-best method by 5.55% and 2.44%, respectively, owing to its robustness in handling the inherent heterogeneity across sites, which is often overlooked by conventional deep learning models.

F. Comparisons of Feature Extractors and Prototype Initialization Strategies

Given the natural graph structure of BFNs, we further compared the transformer with three representative graph feature extractors, including: CNN [48], GCN [49], and Graph transformer [50], where both the GCN and Graph transformer rely on KNN-based adjacency matrices to ensure a fair comparison. As shown in **Supplementary TABLE SI**, our transformer-based backbone consistently achieves the best performance across both datasets, outperforming the CNN that lacks relational modeling, the GCN whose message passing is constrained by a fixed and heuristic adjacency graph, and the Graph transformer that inherits the limitations of predefined graph sparsity. These findings suggest that the transformer's self-attention mechanism provides a more flexible and expressive way to capture long-range functional dependencies

TABLE IV
CLASSIFICATION PERFORMANCE OF OUR METHOD IN ABLATION EXPERIMENTS

Cohort	FBIRN				BSNIP			
Method	ACC	SEN	SPE	AUC	ACC	SEN	SPE	AUC
Ours w/o SMAC	0.7778±0.0521	0.8571±0.0753	0.7083±0.1531	0.8972±0.0404	0.7395±0.0602	0.7188±0.0827	0.8000±0.0407	0.7511±0.0912
Ours w/o SMA	0.8089±0.0199	0.7714±0.1032	0.8417±0.1078	0.8782±0.0306	0.7581±0.0509	0.7813±0.0625	0.6909±0.1220	0.7847±0.0438
Ours w/o SMC	0.8133±0.0199	0.7524±0.1137	0.8667±0.0854	0.8861±0.0208	0.7628±0.0447	0.7500±0.0699	0.8000±0.0407	0.8205±0.0221
Ours w/o SAC	0.8222±0.0272	0.8286±0.0865	0.8167±0.0632	0.8928±0.0271	0.7721±0.0255	0.7562±0.0513	0.8182±0.0909	0.7903±0.0536
Ours w/o MAC	0.8222±0.0351	0.7809±0.1287	0.8583±0.0697	0.9020±0.0319	0.7814±0.0208	0.7750±0.0342	0.8000±0.0407	0.8216±0.0189
Ours w/o SM	0.8266±0.0099	0.7905±0.0426	0.8583±0.0228	0.8897±0.0274	0.7907±0.0285	0.8312±0.0474	0.6728±0.0813	0.7750±0.0333
Ours w/o SA	0.8444±0.0157	0.8953±0.0522	0.8000±0.0348	0.9242±0.0244	0.7907±0.0165	0.8125±0.0585	0.7273±0.1113	0.8170±0.0323
Ours w/o SC	0.8311±0.0511	0.8381±0.0722	0.8250±0.0904	0.9071±0.0266	0.8000±0.0208	0.8000±0.0474	0.8000±0.1494	0.8187±0.0446
Ours w/o MA	0.8311±0.0122	0.8857±0.0543	0.7833±0.0543	0.9123±0.0179	0.8047±0.0312	0.8250±0.0474	0.7455±0.0407	0.8358±0.0156
Ours w/o MC	0.8355±0.0253	0.8667±0.0621	0.8084±0.0959	0.9183±0.0303	0.7954±0.0104	0.8062±0.0559	0.7637±0.1220	0.8313±0.0152
Ours w/o AC	0.8356±0.0337	0.7714±0.0521	0.8916±0.0632	0.9079±0.0458	0.7954±0.0195	0.8188±0.0262	0.7273±0.1286	0.7937±0.0309
Ours w/o S	0.8489±0.0244	0.8571±0.0583	0.8417±0.0618	0.9183±0.0377	0.8186±0.0255	0.8313±0.0171	0.7818±0.0498	0.8352±0.0311
Ours w/o M	0.8444±0.0157	0.8286±0.0543	0.8584±0.0632	0.9127±0.0262	0.8093±0.0104	0.8313±0.0280	0.7455±0.0407	0.8767±0.0470
Ours w/o A	0.8533±0.0199	0.8191±0.1233	0.8833±0.0995	0.9159±0.0296	0.8233±0.0389	0.8438±0.0585	0.7637±0.1651	0.8108±0.0434
Ours w/o C	0.8578±0.0199	0.8957±0.0337	0.8583±0.0475	0.9285±0.0274	0.8140±0.0164	0.8438±0.0313	0.7273±0.0909	0.7977±0.0408
Ours	0.8889±0.0222	0.9143±0.0976	0.8667±0.0457	0.9433±0.0295	0.8605±0.0164	0.8750±0.0221	0.8182±0.0643	0.8392±0.0192

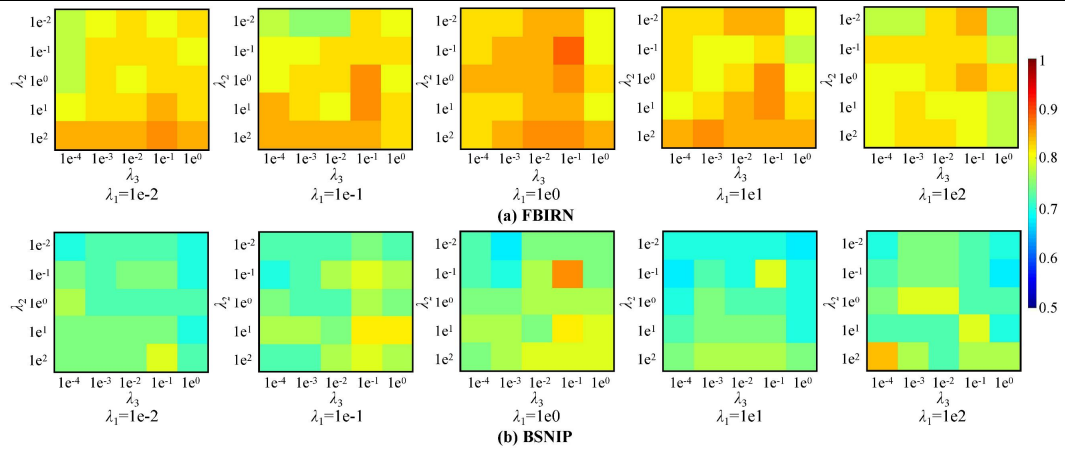


Fig. 3. Classification accuracies with respect to different parameter values.

without being restricted by manually constructed graph structures, thereby producing more discriminative representations for BFN classification. Importantly, experiments across different backbone architectures in **Supplementary TABLE SII–SIII** show that the site-independent learning and prototype learning modules consistently improve performance, demonstrating that the observed gains are not solely attributable to the transformer backbone.

To further ensure that the superiority of our model does not stem from particular prototype initializations, we also compared it with the other two initialization strategies: uniform initialization [51] and Xavier initialization [52]. As reported in **Supplementary TABLE SIV**, the performance variance of the all the three strategies remained very small, with values of 0.0093 on FBIRN and 0.0246 on BSNIP. These results show that our model is robust to different initialization strategies.

G. Ablation Study

To comprehensively evaluate the effectiveness of all the four key components in our total loss function, we conducted an expanded ablation study on both datasets. For clarity, we denoted the $\mathcal{L}_{sil}$, $\mathcal{L}_{mle}$, $\mathcal{L}_{pat}$, and $\mathcal{L}_{pct}$ as S , M , A , and C , respectively. Variants such as Ours w/o MA and Ours w/o MAC indicate the removal of the corresponding components. As shown in **Table IV**, the results exhibited a clear and consistent pattern on two datasets. Each pairwise combination

performed better than using any single component, and combining three components further improved performance compared with all pairwise settings, demonstrating that the components provide complementary benefits rather than redundant effects. The full model that integrated all four components achieved the best overall classification accuracy, indicating that joint optimization enhances model robustness and generalization. Among all components, $\mathcal{L}_{mle}$ was the most critical one, since its removal resulted in the largest performance drop. The second most influential component was $\mathcal{L}_{sil}$, and this finding highlights the importance of stable prototype updating and site-invariant representation learning in multi-site BFN classification.

In addition, to assess the roles of soft assignment and entropy regularization in the Sinkhorn algorithm, we performed an ablation study. Two baseline variants were constructed: the w/o softness variant that replaces soft assignment with a hard nearest-prototype assignment, where each sample is assigned to its closest prototype via an argmax operation and gradients are not propagated through the discrete assignment, and the w/o regularization variant that keeps soft assignment while removing the entropy and marginal regularization imposed by the Sinkhorn algorithm. As shown in **Supplementary TABLE SV**, the full model consistently outperforms both variants. The largest degradation appears in the w/o softness, highlighting the importance of soft assignment in handling uncertainty. The

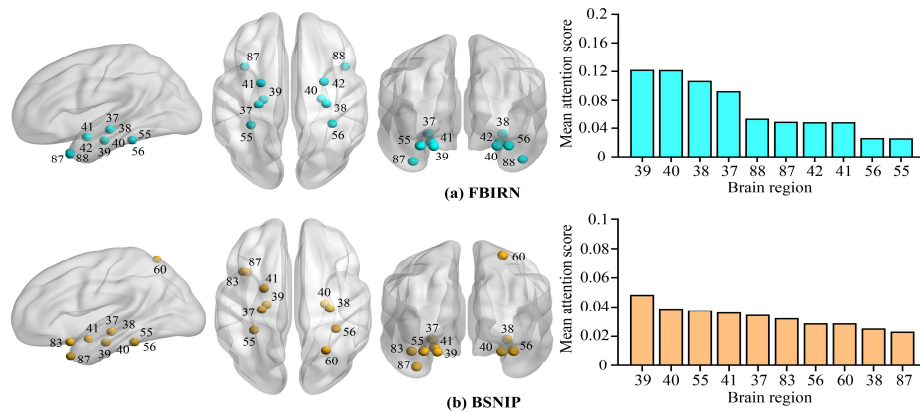


Fig. 4. Visualization of the top 10 most discriminative brain regions on two datasets. (a) FBIRN, (b) BSNIP.

weaker performance of the w/o regularization further demonstrates that the entropy constraint is essential for avoiding prototype collapse and achieving robust generalization.

H. Analysis of Parameter Sensitivity

As indicated in Eq. (11), the total loss function involves three parameters. To assess the impact of these parameters on our experimental outcomes and to identify the optimal values, we utilized a grid search method. Specifically, λ_1 and λ_2 were selected from $\{1e^{-2}, 1e^{-1}, 1e^0, 1e^1, 1e^2\}$, while λ_3 was chosen from the range $\{1e^{-4}, 1e^{-3}, 1e^{-2}, 1e^{-1}, 1e^0\}$. The corresponding results are presented in **Fig. 3**. Overall, our method achieved consistently high ACC on the FBIRN dataset, while exhibiting relatively moderate performance on the BSNIP dataset. The highest ACC was consistently observed on both datasets when $\lambda_1=1e^0$, $\lambda_2=1e^{-1}$, and $\lambda_3=1e^{-1}$, indicating that these parameters generalize well across datasets. We further observed that increasing λ_3 from $1e^{-1}$ to $1e^0$ leads to a clear performance drop, because an overly large HSIC weight can inadvertently weaken class-discriminative information.

In addition, we investigated the effect of varying prototype numbers for the normal and patient classes separately, sweeping from 10 to 100 for each class. As shown in **Supplementary Fig. S1**, the accuracy is consistently high when the patient class uses 50 prototypes, and the highest accuracy is obtained when both the normal and patient classes are set to 50. The patient class is more sensitive to the prototype number due to its higher intra-class heterogeneity. Using around 50 prototypes enables the model to capture diverse patient patterns effectively without redundancy, leading to consistently strong performance across datasets. In contrast, the normal class shows lower variability, suggesting that its features are more consistent and can be adequately captured with a smaller number of prototypes. We also analyzed the influence of the EMA momentum parameter $\alpha \in \{0.5, 0.9, 0.99, 0.995, 0.999, 0.9995\}$ [22]. As observed in **Supplementary Fig. S2**, classification accuracy rises with increasing α , peaks at $\alpha=0.99$ on both datasets, and slightly declines when α becomes larger, indicating that excessively high momentum slows prototype adaptation. Smaller α (e.g., 0.5-0.9) leads to unstable prototype updates and fluctuating contrastive loss, while $\alpha=0.99$ achieves the best trade-off between stability and

adaptability, ensuring robust and consistent performance across datasets. Beyond the parameters discussed above, the entropy regularization coefficient ε directly influences the behavior of the prototype assignment optimization in Eq. (4). We therefore conduct an additional sensitivity analysis on ε , evaluating $\varepsilon \in \{0.05, 0.1, 0.2, 0.3\}$. As shown in **Supplementary Fig. S3**, classification accuracy decreases as ε increases, indicating that overly strong entropy regularization harms performance. Larger ε values yield more uniform prototype assignments, weakening class-specific alignment and reducing discriminative capacity. Accordingly, $\varepsilon=0.05$ provides the best balance between assignment smoothness and discriminative learning.

I. Discriminative Brain Regions

We visualized and ranked the average attention weights assigned by the transformer encoder across all target domain samples, highlighting the top 10 most discriminative brain regions contributing to the model's prediction. The top 10 SZ-related brain regions were visualized using BrainNet Viewer toolbox, with their average attention weights shown in **Fig. 4**. Temporal lobe regions were identified across both datasets. In FBIRN, left parahippocampal gyrus (PHG.L), right parahippocampal gyrus (PHG.R), left hippocampus (HIP.L), right hippocampus (HIP.R), left temporal pole: middle temporal gyrus (TPOMid.L), right temporal pole: middle temporal gyrus (TPOMid.R), left amygdala (AMYG.L), right amygdala (AMYG.R), left fusiform gyrus (FFG.L) and right fusiform gyrus (FFG.R) were identified, while BSNIP detected PHG.L, PHG.R, FFG.L, AMYG.L, HIP.L, left temporal pole: superior temporal gyrus (TPOsup.L), FFG.R, HIP.R, and TPOMid.L. Notably, PHG.L, PHG.R, HIP.L, HIP.R, TPOMid.L, AMYG.L, FFG.L and FFG.R were consistently identified in both datasets.

IV. DISCUSSION

In this paper, we proposed a novel DG framework for multi-site BFN classification, consisting of three key components: (1) a transformer encoder was used to extract discriminative subject-level features from BFNs, (2) the extracted features were projected through a site-independence module, which employs a nonlinear projection regularized by HSIC to mitigate site-specific biases, and (3) prototype learning

was performed by softly assigning projected features to class-specific prototypes via Sinkhorn. Prototypes were updated via EMA, while the MLE loss optimized the matching probabilities between features and prototypes in the probabilistic space. Contrastive and alignment losses further enhanced inter-class separation and intra-class consistency. Experimental results on two SZ datasets demonstrated that our method outperformed 7 DG, 6 DA, 6 multi-site and 6 state-of-the-art methods. This method also enabled identification of discriminative brain regions in unseen sites, demonstrating its potential for discovering generalizable biomarkers.

A major contribution of this work is the integration of site-invariant feature learning and prototype-based modeling into a unified framework, enabling the extraction of domain-robust and class-discriminative representations. While most existing DG methods rely on meta-learning or global distribution alignment to simulate domain shifts [10-12], they often involve complex nested optimization and fail to explicitly model intra-class diversity. Such limitations hinder their ability to generalize in highly heterogeneous multi-site scenarios. In contrast, our method jointly learns site-invariant and class-discriminative features by incorporating HSIC-regularized projection and prototype-based learning within a unified framework. By modeling intra-class structural variations, the model enhances feature discriminability and improves generalization across domains in a unified training process. As shown in **Table I**, our method consistently outperformed other DG methods across both datasets.

In addition, we conducted comparative experiments with representative DA and multi-site related methods to further validate the effectiveness of our method. It is important to note that DA methods operate under a more favorable setting, as they require access to unlabeled target domain data during training, a condition that is often unrealistic in clinical practice. Multi-site methods often rely on handcrafted harmonization procedures or complex graph construction strategies that increase model complexity and reduce scalability. In contrast, our framework followed the more challenging DG setting, where no target data or explicit site correction was available. Despite this stricter assumption, our method still achieved competitive performance, highlighting its robustness and practical applicability in realistic multi-site scenarios.

Besides improving classification performance, our method also revealed discriminative brain regions that are consistent with findings from existing studies. Temporal lobe regions were consistently identified across both the FBIRN and BSNIP datasets, including the PHG.L/R, HIP.L/R, TPOMid.L, AMYG.L, and FFG.L/R. These regions have been frequently reported as key biomarkers for SZ diagnosis in previous studies. Specifically, the HIP and AMYG are associated with impaired memory and emotional regulation in SZ [53], while the PHG and FFG have been linked to perceptual and cognitive dysfunction [54]. These findings highlight the consistency and biological relevance of the brain regions identified by our model across multi-site datasets.

Our method also has several limitations. First, the

computational complexity of our method is $O(LB(N^2d + Nd^2) + Bd^2 + B^2d + Bpd + SBp + P^2d)$, which exceeds that of other DG methods that scale more linearly with data size. This results in a longer training time (64.88s versus 26.27-53.42s for other methods), mainly due to the self-attention cost and iterative Sinkhorn updates. Future work may reduce this overhead through more efficient attention mechanisms or lightweight prototype learning. Second, the site-invariant projection relies solely on HSIC to measure feature-site dependence. Although effective, it may not capture more subtle nonlinear or hierarchical relationships. Incorporating alternative dependence metrics, such as distance correlation or kernel-based mutual information, may improve robustness. Third, although the transformer encoder models global brain region interactions, its attention mechanism treats all pairwise connections equally, which may introduce noise from irrelevant regions. Introducing anatomical priors or attention gating could help focus on more meaningful connections and reduce unnecessary complexity.

V. CONCLUSION

In this paper, we proposed a novel DG framework for multi-site BFN classification, which integrated a transformer encoder for extracting global subject-level representations, a site-independence module for reducing site-specific biases, and a prototype learning module for capturing intra-class diversity through multiple class-specific representations. The effectiveness of our method was validated on SZ classification using two independent datasets. Our method achieved the highest ACC compared with 7 DG, 6 DA, 6 multi-site and 6 state-of-the-art methods. Collectively, the proposed method effectively enhances generalization across sites, improves class-level discriminability, and identifies consistent disorder-related brain regions, demonstrating its potential for robust and interpretable neuroimaging-based diagnosis.

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